The Hendrickson approach, the hybrid design, and when component decomposition pays off

2026-06-17 10:40 PDT

> **SUPERSEDED (2026-09-09). Do not cite or build on this document.** Its content is merged into `docs/31-component-decomposition-and-reductions.md`, which also adds Part IV on the two-component reductions (papers 13 and 14). This copy is retained for history only and received a light review on 2026-10-07 (notation, obvious errors, US English); its numbers were not re-verified. Consult docs/31 instead.

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Purpose. Historical record of the June 2026 synthesis of the original Hendrickson approach, the identifiability the hybrid design supplies, and the paper-06 conclusion on when component decomposition repays its cost. Superseded by docs/31.

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Notation index and glossary

Notation follows `analysis/report/NOTATION.md`. Code identifiers appear in backticks.

Table 1: Notation index and glossary of symbols used in this document

Symbol	Meaning	Status
BR, PB, TV	biological, placebo-belief and time-variant response components	canonical
Y	observed symptom score (code `Sx`)	canonical
BL_i	participant baseline level	canonical
η_t	design expectancy at visit t: 1 open label, 0.5 blinded, 0 covert (code `e`, `expectancies`)	canonical
D_{bc}	exposure-decayed drug indicator (code `Dbc`)	canonical
λ, $t_{1/2}$, t_{sd}	carryover decay rate $\ln 2/t_{1/2}$; half-life; time since discontinuation	canonical
$\beta_{bm}^{\text{lumped}}$, β_{bm}^{BR}, β_{bm}^{PB}, β_{bm}^{TV}	one-component and component-specific biomarker slopes	canonical
w_{PB}, w_{TV}	design-dependent weights in the omitted-variable-bias identity	local
m_{BR}, m_{PB}, m_{TV}	Gompertz maxima of the three components (code `max`)	local
r_{BR}, d_{BR}	Gompertz rate and displacement of BR (code `rate`, `disp`)	local
$c_{bm,PB}$	biomarker-PB coupling correlation in the contamination studies	local
N	total sample size	canonical
α	significance level	canonical

- **One-component (lumped) analysis.** The Hendrickson model with a single drug indicator, `Sx ~ bm + t + Dbc + bm:Dbc`.

- **Attribution.** Apportioning response among pharmacology (BR), belief (PB) and natural history (TV).

1. Purpose and scope

This white paper consolidates, in one place, three things that are otherwise distributed across the package source and the paper-06 manuscript (`analysis/report/06-component-decomposition/`):

1. what the original Hendrickson et al. (2020) approach actually does, on both the data-generating and the analysis side;
2. how the hybrid open-label-plus-blinded-discontinuation design supplies identifiability, and what it cannot supply;
3. the conclusion the paper-06 analysis draws about when component decomposition repays its cost, with explicit, practical advice on how and when to field the hybrid design.

The intended reader is an investigator deciding whether to run an aggregated N-of-1 trial under the hybrid design and whether to layer a component decomposition on top of the standard lumped analysis. The treatment is academic and self-contained; the simulation numbers are quoted from the realized Study A, the Study B pilot, and Study C reported in the paper-06 manuscript.

2. The original Hendrickson approach

The founding pmsimstats question is the detection of a biomarker-by-treatment interaction in an aggregated N-of-1 trial. The Hendrickson reference implementation (`implementations/original/`) is the historical anchor for the whole program. Its defining feature, and the starting point for paper 06, is an asymmetry between how it generates data and how it analyzes it.

2.1 The data-generating process is fully decomposed

The DGP in `implementations/original/R/generateData.R` builds the observed symptom score as the sum of three causally distinct components, each a modified Gompertz trajectory with participant-specific random effects:

- BR (biological response, the pharmacological component), driven by cumulative time on drug, decaying off drug with a carryover half-life;
- PB (placebo-belief response), driven by cumulative time on belief and scaled by the phase expectancy η_t;
- TV (time-variant natural-history component), driven by time since trial entry.

The components are summed and subtracted from baseline (lines near 99 and 104 of `generateData.R`):

```
D_t  = rowSums(t.br + t.pb + t.tv)    # total response at timepoint t
Sx_t = BL - D_t                        # symptom = baseline minus total
```

The raw simulated data retain the individual component values (`t1.br`, `t1.pb`, `t1.tv`, and so on for each timepoint), so the ground truth is three-component by construction. The prazosin-PTSD calibration used throughout the program draws its Gompertz parameters from the Hendrickson et al. (2020) dataset ($m_{BR} \approx 11$, $r_{BR} \approx 0.42$ per week, $d_{BR} \approx 5$).

2.2 The analysis model is a single lumped drug indicator

The analysis in `implementations/original/R/lme_analysis.R` does not mirror the DGP. It fits one continuous drug indicator, not three components:

```
Sx ~ bm + t + Dbc + bm:Dbc
random      = ~ 1 | ptID
correlation = corCAR1(form = ~ t | ptID)
```

where `Dbc` is the continuous drug state D_{bc} (equal to 1 on drug and to $e^{-\lambda t_{sd}}$ off drug, with $\lambda = \ln 2 / t_{1/2}$), and `bm:Dbc` is the biomarker-by-drug interaction. Variant forms in the same file add an expectancy indicator (`De`) or a carryover term (`tsd`), but they remain single-drug-indicator models. The model estimates one lumped drug main effect and one lumped biomarker slope; it does not recover BR, PB, and TV separately.

2.3 The asymmetry that paper 06 studies

The original therefore applies decomposition only as a data-generating device. It knows the response is $BR + PB + TV$ and simulates accordingly, but its inferential model collapses the three into a single treatment coefficient. This lumped estimator is exactly the baseline that paper 06 formalizes, and the central question of that paper is the consequence of the asymmetry: when does the lumped analysis of three-component data give a biased or uninformative answer, and when does adding analysis-side decomposition fix it?

3. The hybrid design

Component decomposition is meaningless unless the data contain enough information to pin down each component separately. That information is supplied by trial design, not by the analysis model. The hybrid open-label-plus-blinded-discontinuation design is the worked example throughout paper 06 and the primary identifiability vehicle.

3.1 Phase structure

The hybrid design strings together three phases, each contributing a different mixture of the three components:

Table 2: Phases of the Hybrid design, with drug state, expectancy and the response components present in each

Phase	Drug	Belief	Components present
Open-label on drug	Yes	Full ($\eta_t = 1$)	$BR + PB(\eta_t = 1) + TV$
Blinded discontinuation	Hidden	Hedged ($\eta_t = 0.5$)	BR (on drug) $+PB(\eta_t = 0.5) + TV$
Blinded crossover	Either	Hedged ($\eta_t = 0.5$)	BR (on drug) $+PB(\eta_t = 0.5) + TV$

The 'on drug' qualifier on BR is shorthand for the continuous-decay drug state D_{bc}, not a binary switch. (*Correction, 2026-10-07:* the third row previously read 'Open-label crossover, knows, $\eta = 1$'. In the Hybrid design as coded, the expectancy vector is `c(1, 1, 0.5, 0.5, 0.5, 0.5, 0.5, 0.5)`, so the crossover is blinded with $\eta_t = 0.5$; inspected in `analysis/scripts/carryover-sensitivity/simulation-core.R` and the `figure4` drivers. The same row in docs/31 needs the same fix.)

3.2 The three identifying contrasts

Three phase-by-allocation contrasts identify the three components:

- **Open-label versus blinded (within drug)** yields $PB(\eta_t = 1) - PB(\eta_t = 0.5)$, which is approximately $0.5\,PB$, and so isolates the belief component, conditional on a known blinded-phase expectancy.
- **Blinded on-drug versus blinded placebo** yields BR once drug carryover has elapsed, because PB and TV match across allocation arms within the blinded phase.
- **Off-drug phases** yield TV plus any residual off-drug PB, with no active BR contribution beyond carryover.

The lumped LME combines these into one model, but the inference is driven by the design. The one-component analysis already exploits the drug-on/off contrast through its `Dbc` term; that is why, as Section 5 shows, it is not as impoverished as its single coefficient suggests.

3.3 The expectancy (η) identification limit

The hybrid design has one principled limitation that deserves emphasis. With only an open-label and a blinded phase, the data identify the product $\eta\, m_{PB}$, not the two factors separately. The working value η approximately 0.5 in the blinded phase is a modeling assumption, not an estimate, and absent a belief measurement it is not falsifiable from outcome data alone. The identified set for m_{PB} is the interval obtained by dividing the estimated product by η over its plausible range, collapsing to a point only when η is pinned down. Two design features close the gap:

- an open-label placebo phase, which supplies a second equation that separates η from m_{PB};
- direct measurement of treatment expectancy, as in the open-hidden and balanced-placebo designs that manipulate or record belief.

Any trial relying on the decomposition for a PB-versus-BR attribution should include such a phase or collect an expectancy measure, and should report sensitivity of the attribution to η across its plausible range.

3.4 Identifiability is necessary but not sufficient

Design-contrast identifiability does not guarantee stable estimation at finite sample sizes. The three Gompertz trajectories share an onset shape and overlap in calendar time, so their parameters can be near-collinear even when the identifying contrasts are present. Joint recovery additionally requires a phase in which each component varies while the others are held fixed, and an off-drug window long enough, relative to the carryover half-life, for TV to separate from a still decaying BR. Where these conditions are weak the components are identified only up to a near-collinear direction; the symptom is the rank deficiency the Study B pilot encountered when the full phase-augmented formula was fitted at $N = 35$. Recoverability is to be mapped by identifiability diagnostics (convergence rates, variance-inflation factors on the component-specific parameters, and sensitivity of the BR estimate to the assumed η), not asserted.

4. The paper-06 analysis

4.1 The omitted-variable-bias identity

The analytic core of paper 06 is an exact identity for what the lumped estimator estimates. By linearity of covariance, the lumped biomarker-by-treatment slope decomposes as

$$\beta_{bm}^{\text{lumped}} = \beta_{bm}^{BR} + w_{PB}\,\beta_{bm}^{PB} + w_{TV}\,\beta_{bm}^{TV}$$

so the lumped slope is displaced from the pharmacological slope β_{bm}^{BR} by a term proportional to the biomarker's covariance with PB and TV, weighted by design-dependent weights w_{PB} and w_{TV}. The displacement is in the probability limit, not the variance: a larger one-component trial converges on the same biased value. The bias is zero if and only if the biomarker is orthogonal to PB and TV, regardless of the magnitude of those components. The companion identity for the lumped treatment effect is displaced by the PB and TV contributions themselves, so a lumped pre-post treatment effect is biased whenever PB or TV is non-zero.

4.2 Three analysis strategies compared

Paper 06 compares three analysis strategies against three-component data:

1. **One-component (lumped).** The Hendrickson model, `Sx ~ bm + t + Dbc + bm:Dbc`.
2. **Phase-augmented LME.** A modest extension, `Sx ~ bm + t + Dbc + phase + Dbc:phase + bm:Dbc + bm:Dbc:phase`, which tests whether the drug and biomarker-by-drug effects attenuate under blinding, without the full Gompertz parametric form.
3. **Full component-matched decomposition.** A parametric model that mirrors the DGP and recovers component-specific slopes, feasible only inside the recoverable region of Section 3.4.

4.3 What the simulations show

Study A (orthogonal-biomarker arm, hybrid design; 1,000 replicates per alternative cell, 5,000 per null). With the biomarker coupled to BR only ($c_{bm,PB} = 0$), the one-component estimate of the biomarker-by-treatment slope is essentially unbiased across the entire (m_{PB}, m_{TV}) grid: mean absolute bias 0.016 at $N = 35$ and 0.006 at $N = 150$ against a true value of -2.25, both within one Monte Carlo standard error of zero, with near-nominal type I error (0.031 to 0.056 for the one-component analysis). The phase-augmented analysis reduces no bias (0.028 versus 0.016 at $N = 35$) and is markedly less powerful (mean power 0.35 for the phase-augmented analysis versus 0.59 for the one-component analysis at $N = 35$; 0.62 versus 0.90 at $N = 70$; both reach approximately 1.0 by $N = 100$; order of the pairs corrected 2026-10-07 against paper 06, `report-slim.Rmd`). Under an uncontaminated biomarker, decomposition is therefore strictly dominated.

Study B (coupling-arm pilot; 100 replicates per cell). With the biomarker coupled to PB, the one-component bias scales with the coupling strength $c_{bm,PB}$, not with the magnitude of PB, exactly as the identity predicts

(the one-component bias is approximately 0.51 at the contamination level studied). The phase-augmented analysis offers no protection: its bias is statistically indistinguishable from the one-component bias at every contamination level.

Study C (recovery under a belief-decoupling design; 1,000 replicates per cell, N in {70, 150}, $c_{bm,PB}$ in {0, 0.1, 0.2, 0.3}). This is the decisive design comparison. On the standard hybrid design the drug-state and belief contrasts are collinear (analysis-stage correlation approximately +0.6), and the belief-covariate, blinded-stratum, and Gompertz-basis decompositions all fail to recover β_{bm}^{BR}, no better than the one-component analysis. A balanced-placebo (open-hidden) design adds a covert on-drug phase ($\eta_t = 0$) and an open-placebo phase ($\eta_t = 1$), reducing the drug-state-by-belief correlation to approximately -0.66 and breaking the collinearity. On that design the belief-covariate decomposition returns the unbiased pharmacological slope across the feasible contamination range (bias -0.022, +0.008, +0.021, -0.026 at $N = 150$, MCSE approximately 0.02; coverage 0.94 to 0.96), whereas the one-component estimator degrades monotonically (bias +0.16, +0.34, +0.48; coverage down to 0.86). The grid is capped at $c_{bm,PB} = 0.30$ because beyond approximately 0.45 the implied covariance is non-positive-definite.

The power comparison between the two estimators on the balanced-placebo design is more nuanced than a uniform decomposition penalty (rejection rate for the interaction at $\alpha = 0.05$, computed from the 1,000-replicate per-cell p-values; paired standard error of the gap at most 0.014):

Table 3: Power for the interaction on the balanced-placebo design, one-component versus belief-covariate decomposition (Study C, 1,000 replicates per cell)

$c_{bm,PB}$	N	One-component	Decomposition	Gap
0.0	70	0.85	0.66	+0.19
0.1	70	0.78	0.67	+0.11
0.2	70	0.74	0.69	+0.05
0.3	70	0.64	0.67	-0.03
0.0	150	0.99	0.93	+0.07
0.1	150	0.98	0.93	+0.05
0.2	150	0.97	0.93	+0.04
0.3	150	0.93	0.93	-0.00

At zero contamination the one-component analysis is the more powerful of the two (by 0.19 at $N = 70$, 0.07 at $N = 150$), reflecting the degrees of freedom the decomposition spends. That advantage erodes monotonically as contamination grows and reverses by $c_{bm,PB} = 0.3$ (0.64 versus 0.67 at $N = 70$), because contamination in this direction drags the biased one-component estimate toward the null and so costs it power, while the decomposition holds the estimate at the pharmacological target. The one-component power advantage is therefore

confined to the low-contamination regime in which decomposition is unnecessary. (The sign of the gap depends on the direction of the coupling: a coupling that reinforced the BR sign would inflate the one-component estimate and give it spuriously higher power, which is power to detect a contaminated quantity rather than a genuine advantage.)

On the hybrid design the same comparison is uninformative because power saturates near 1.0 for both estimators (the BR effect of -2.25 is large relative to its standard error at $N = 70$ to 150), and both are biased under contamination in any case; there the operative axis is bias and coverage, not power.

5. The conclusion drawn from the 06 analysis

The inferential value of decomposition is conditional, not universal, and the condition is a joint function of the biomarker's correlation structure and the trial design, not of either alone. Three points summarize the result.

1. **The design contrast matters, not the parametric model.** The trial must supply the drug-on/off and belief contrasts that identify the components. The lumped one-component analysis already exploits the drug-on/off contrast through `Dbc`. Adding parametric component structure on top is not free: it spends degrees of freedom and, under an uncontaminated biomarker, reduces no bias while losing substantial power.
2. **For the biomarker-by-treatment slope, the bias is governed by covariance, not magnitude.** A biomarker orthogonal to PB and TV gives an unbiased lumped slope no matter how large PB and TV are. A biomarker coupled to PB or TV gives a contaminated slope that no amount of data fixes, because the failure is identification, not precision.
3. **Recovery under contamination is a design property.** When the biomarker is coupled to PB, the bias is recoverable only under a design that varies belief independently of drug exposure. On the standard hybrid design no analysis recovers it; on a balanced-placebo design the belief-covariate decomposition does.

The bottom line specific to the hybrid design

For the biomarker-by-treatment interaction estimand on the hybrid design, component decomposition never wins:

- if the biomarker is orthogonal to PB and TV, the lumped analysis is already unbiased and is the more powerful of the two, so decomposition only costs power;

- if the biomarker is coupled to PB, the lumped analysis is biased but decomposition cannot rescue it on the hybrid design, because the identifying contrasts are collinear.

Decomposition earns its place on the hybrid design only for a different estimand: the **attribution** of response to pharmacology versus belief versus natural history. For attribution the hybrid design is the primary identification vehicle, subject to the η limit of Section 3.3 and the collinearity caveat of Section 3.4.

Was the decomposition in Hendrickson's DGP unnecessary?

A natural reading of the paper-06 result is that, because the lumped analysis suffices for the biomarker-by-treatment estimand, the three-component structure Hendrickson built into the simulator was itself superfluous. That reading conflates two distinct choices. The decomposition was unnecessary in the *analysis* model; it was load-bearing in the *data-generating process*. The two claims are not interchangeable.

The DGP decomposition was necessary, or at least not removable, for two reasons.

- It defines the ground truth the whole program tests against. The omitted-variable-bias identity of Section 4.1, and every simulation that confirms it, can be exhibited only if the simulator generates BR, PB, and TV as separately controllable channels. A DGP that produced a single lumped treatment effect would offer no $c_{bm,PB}$ knob to turn, no way to construct the contamination, and therefore no way to show that the lumped analysis is unbiased under orthogonality but biased under coupling. The DGP decomposition is what makes the bias question askable.
- It encodes genuinely distinct mechanisms with different time-drivers: BR by cumulative time on drug with off-drug carryover decay, PB by time on belief scaled by the phase expectancy η, and TV by time since entry. These produce different trajectories across trial phases, and the design-contrast structure of Section 3.2 (open-label versus blinded yielding approximately $0.5\,PB$, blinded on-drug versus blinded placebo yielding BR) cannot be reproduced from a single lumped curve. The components are not cosmetic; they drive the phase behavior the whole identification argument relies on.

What paper 06 shows to be unnecessary is only the symmetric move of mirroring that structure in the analysis. Fitting the components back out is unnecessary, and under an orthogonal biomarker counterproductive, for the biomarker-by-treatment estimand, and unrecoverable on the hybrid design when the biomarker is coupled. Hendrickson's choice to analyze with a single lumped `Dbc` was, in hindsight, the appropriate one for that estimand.

The qualifier is the estimand. The analysis-side lumping is validated as sufficient only for the interaction-slope target under an orthogonal biomarker. Had the scientific target been attribution, or a PB-coupled biomarker, the lumped analysis would be insufficient, and the DGP's decomposition would then require both a matching analysis and a belief-decoupling design. The accurate summary is therefore that the DGP decomposition was necessary to pose and answer the question, whereas the analysis-side decomposition was unnecessary for the specific estimand Hendrickson actually pursued.

6. Advice: how and when to use the hybrid design

The hybrid design is the right default for aggregated N-of-1 trials whose primary estimand is the pharmacological treatment effect or a biomarker-by-treatment interaction with a mechanistically pharmacological biomarker. It is not the right design when the biomarker may couple to placebo responsiveness and the pharmacological slope must be recovered. The following guidance operationalizes that.

Use the hybrid design, with the lumped one-component analysis, when:

- the estimand is the lumped treatment effect or a biomarker-by-drug interaction, and
- the candidate biomarker is mechanistically orthogonal to belief and to natural history. Pharmacokinetic or receptor-binding biomarkers (for example a CYP metabolizer phenotype, or an adrenergic-receptor variant affecting prazosin binding) qualify: enzymatic metabolism and receptor affinity cannot plausibly couple to belief state or untreated disease trajectory. Here the lumped slope is already the target, and a parametric component model is counterproductive.

Use the hybrid design, but add an open-label placebo phase or an expectancy measure, when:

- the estimand is attribution (how much of the response is pharmacological versus belief versus natural history). The base two-phase hybrid identifies only the product $\eta\, m_{PB}$; the open-label placebo phase or a recorded expectancy separates them. Report attribution sensitivity to η across its plausible range, and verify the off-drug window is long enough, relative to the carryover half-life, for TV to separate from a decaying BR.

Do not rely on the hybrid design; field a balanced-placebo (open-hidden) design instead, when:

- the estimand is the pharmacological biomarker slope and the candidate biomarker may correlate with placebo responsiveness or with natural-history drift. Psychological or symptom-severity biomarkers fall here: baseline PCL-5 severity, for example, couples to both expectancy and regression-to-the-mean TV, and a COMT genotype is a documented predictor of placebo-response magnitude. On the hybrid design the contamination is unrecoverable; only the covert-drug and open-placebo phases of the balanced-placebo design decouple belief from drug exposure enough to recover the unbiased slope.

Design-stage checklist before committing to the hybrid design.

- Identify the estimand first: lumped effect, attribution, or pharmacological biomarker slope. The estimand, not convenience, selects the design.
- Classify the biomarker as orthogonal to or potentially coupled with PB and TV, on mechanistic grounds, before enrollment. If coupling cannot be ruled out, the hybrid design is the wrong instrument.
- Verify that every component the estimand requires has at least one phase or contrast in which it is uniquely present. A pure open-label design cannot separate BR from PB; a pure blinded design cannot reach full-strength PB.
- If attribution is in scope, plan the open-label placebo phase or expectancy measurement at design time, and size the off-drug window against the carryover half-life.
- Do not plan to buy bias protection with the analysis model. On the hybrid design the phase-augmented and full-decomposition analyses do not reduce biomarker-slope bias; design is the only lever.

7. Limitations and open items

The Study C recovery result is established at 1,000 replicates over $c_{bm,PB}$ in $\{0, 0.1, 0.2, 0.3\}$ and N in $\{70, 150\}$; recovery beyond the positive-definite range ($c_{bm,PB}$ above approximately 0.45), participant-level component recovery, and replication under an mean-moderation coupling remain open. The Study B coupling-arm evidence is a 100-replicate pilot; its production run is pre-registered. The η working value of 0.5 in the blinded phase remains a modeling assumption absent a belief measurement. The framework's participant-level claims are conditional on falling inside the recoverable region of Section 3.4.

References

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